

Design and Implementation of a Database Management System for Restaurant Table Reservation and Food Service Management System

A DBMS Project

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Problem Statement

Current Challenges

- Double-booking conflicts
- Real-time availability tracking
- Scattered customer records
- Unstructured order management
- Billing inconsistencies
- Data integrity issues

Proposed Solution

- Centralized database system
- Prevent conflicting reservations
- Efficient table management
- Integrated food ordering
- Automated billing
- Data consistency enforcement

Proposed System & Objectives

System Purpose

Structured database system for restaurant operations

Key Functionalities

- View available tables & time slots
- Make & manage reservations
- Browse restaurant menu
- Place & track food orders
- Generate billing details
- Maintain customer information
- Prevent reservation conflicts

Database Role

- Centralized data storage
- Enforces data integrity
- Supports efficient queries
- Maintains data relationships

System Entities

Our database system is built around six major entities that manage reservations, dining, and customer data.

Customer

Stores customer information and tracks reservations

Restaurant Table

Represents physical tables with capacity and status

Reservation

Links customers to tables for specific dates/times

Menu Item

Catalog of food items with pricing and availability

Order

Food order placed during a dining session

Order Item

Individual items within an order with quantity

Schema Relationships & Data Integrity

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Foreign Key Relationships

- CUSTOMER.customer_id → RESERVATION.customer_id
- RESTAURANT_TABLE.table_id → RESERVATION.table_id
- RESERVATION.reservation_id → ORDER.reservation_id
- ORDER.order_id → ORDER_ITEM.order_id
- MENU_ITEM.item_id → ORDER_ITEM.item_id

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Data Integrity Constraints

- Referential Integrity: Foreign keys ensure valid references
- Unique Constraints: Prevent duplicate date/time/table combinations
- Primary Key Constraints: Unique identifiers for each entity
- Domain Constraints: Appropriate data types and value ranges

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Normalization

- Normalized to 3NF (Third Normal Form)
- Eliminates data redundancy
- Ensures efficient query performance
- Maintains consistency

Conclusion

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Summary

Comprehensive database system for restaurant table reservation and food service management addressing key operational challenges.

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Key Achievements

- Identified six core entities
- Developed clear Entity-Relationship Diagram
- Converted to normalized relational schema
- Ensured data integrity through constraints
- Prevented reservation conflicts

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System Benefits

- Centralized data storage
- Efficient queries and reporting
- Data consistency enforcement
- Scalability for future enhancements

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Next Steps

- SQL implementation
- User interface development
- Query optimization
- Deployment and maintenance